Signals & Systems

Assignments 3 & 4 — Complete Worked Solutions

Fourier Series • Fourier Transform

FAST–NUCES, Lahore · EE2023 / EE2008 · Spring 2024
Assignments by Dr. Sajid

How to use this document. Every problem is solved from first principles with the intermediate steps shown, and each answer that has a spectrum is accompanied by a magnitude/phase plot. A handful of problems specify the signal only through a sketch in the original PDF; for those, the waveform actually used is stated explicitly in an orange box. If your printed figure differs, the *method* is identical—only the numbers in the final formula change, and the box tells you exactly which assumption to revisit.

Contents

Conventions used throughout

For a periodic signal of period T_0 and fundamental frequency $\omega_0 = 2\pi/T_0$:

$$\textbf{Exponential FS:}\quad x(t) = \sum_{n=-\infty}^{\infty} D_n\, e^{jn\omega_0 t}, \qquad D_n = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-jn\omega_0 t}\, dt.$$

$$\textbf{Compact trigonometric FS:}\quad x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos\big(n\omega_0 t + \theta_n\big).$$

The two are linked by

$$C_0 = D_0, \qquad C_n = 2\,|D_n|\ (n \geq 1), \qquad \theta_n = \angle D_n, \qquad D_{-n} = D_n^*\ \text{(real signals)}.$$

For the Fourier transform we use the (Lathi) convention

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\, \mathrm{e}^{-\mathrm{j}\omega t}\, \mathrm{d}t, \qquad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, \mathrm{e}^{\mathrm{j}\omega t}\, \mathrm{d}\omega,$$

with $\operatorname{sinc}(x) = \dfrac{\sin x}{x}$ and the key pair $\operatorname{sinc}(at) \overset{\mathcal{F}}{\longleftrightarrow} \dfrac{\pi}{a}\operatorname{rect}\big(\frac{\omega}{2a}\big)$ (a gate of height π/a for $|\omega| < a$).

1 Assignment 3 — Fourier Series

1.1 Q1 — Full-wave rectified cosine

The signal is $x(t) = A\,|\cos \pi t|$. Because $\cos \pi t$ has period 2, taking the absolute value halves it, so

$$T_0 = 1, \qquad \omega_0 = \frac{2\pi}{T_0} = 2\pi.$$

$x(t)$ is **even**, hence $b_n = 0$ and only cosine terms survive. On the interval $\big(-\frac{1}{2}, \frac{1}{2}\big)$ the cosine is non-negative, so $|\cos \pi t| = \cos \pi t$ there.

DC term.

$$C_0 = a_0 = \frac{1}{T_0}\int_{-1/2}^{1/2} A\cos \pi t\, \mathrm{d}t = A\Big[\tfrac{\sin \pi t}{\pi}\Big]_{-1/2}^{1/2} = \frac{A}{\pi}\big(1 - (-1)\big) = \boxed{\frac{2A}{\pi}}.$$

Harmonics. Using $\cos \pi t \cos 2\pi n t = \frac{1}{2}\big[\cos(2n+1)\pi t + \cos(2n-1)\pi t\big]$,

$$a_n = 2A\int_{-1/2}^{1/2} \cos \pi t \cos 2\pi n t\, \mathrm{d}t = A\left[\frac{\sin(2n{+}1)\pi t}{(2n{+}1)\pi} + \frac{\sin(2n{-}1)\pi t}{(2n{-}1)\pi}\right]_{-1/2}^{1/2} = \frac{4A\,(-1)^{n+1}}{\pi\,(4n^2-1)}.$$

The magnitude is $\dfrac{4A}{\pi(4n^2-1)}$ (positive, since $4n^2 - 1 > 0$); the sign $(-1)^{n+1}$ supplies the phase: $\theta_n = 0$ for odd n and $\theta_n = \pi$ for even n.

Result

$$x(t) = \frac{2A}{\pi} + \frac{4A}{\pi}\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{4n^2-1}\cos(2\pi n t), \qquad C_1 = \frac{4A}{3\pi} \approx 0.424A,\;\; C_2 = \frac{4A}{15\pi},\;\; C_3 = \frac{4A}{35\pi}.$$

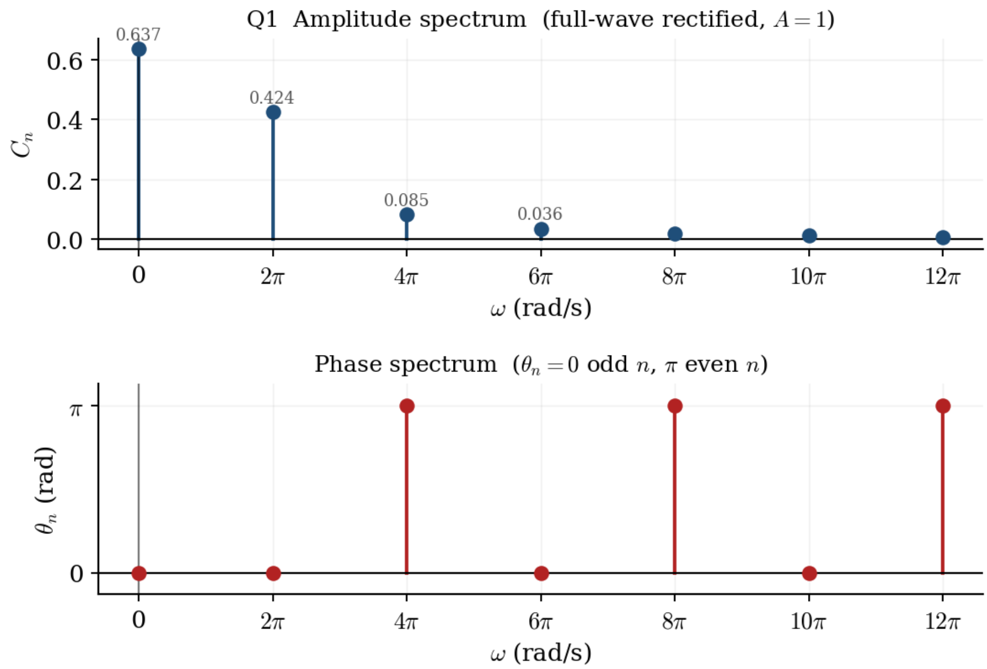

Figure 1: Q1: amplitude and phase spectra of the full-wave rectified cosine ($A = 1$). Lines fall off as $1/(4n^2 - 1)$; the phase alternates $0, \pi$.

1.2 Q2 — Triangular wave

The triangle peaks at A at $t = nT_0$ and crosses zero at $t = (n + \frac{1}{2})T_0$, so on one period $x(t) = A\left(1 - \frac{2|t|}{T_0}\right)$ for $|t| \le T_0/2$. It is **even**, so $b_n = 0$.

$$C_0 = a_0 = \frac{2}{T_0}\int_0^{T_0/2} A\left(1 - \tfrac{2t}{T_0}\right) \mathrm{d}t = \frac{2A}{T_0}\left[t - \tfrac{t^2}{T_0}\right]_0^{T_0/2} = \frac{2A}{T_0}\cdot\frac{T_0}{4} = \boxed{\frac{A}{2}}.$$

Integrating $a_n = \dfrac{4A}{T_0}\displaystyle\int_0^{T_0/2}\left(1 - \tfrac{2t}{T_0}\right)\cos(n\omega_0 t)\,\mathrm{d}t$ by parts gives the standard triangular result

$$a_n = \frac{2A\big(1 - (-1)^n\big)}{\pi^2 n^2} = \begin{cases} \dfrac{4A}{\pi^2 n^2}, & n \text{ odd}, \\ 0, & n \text{ even}, \end{cases} \qquad \theta_n = 0.$$

Result

$$x(t) = \frac{A}{2} + \frac{4A}{\pi^2}\left[\cos\omega_0 t + \frac{1}{9}\cos 3\omega_0 t + \frac{1}{25}\cos 5\omega_0 t + \cdots\right], \qquad \omega_0 = \frac{2\pi}{T_0}.$$

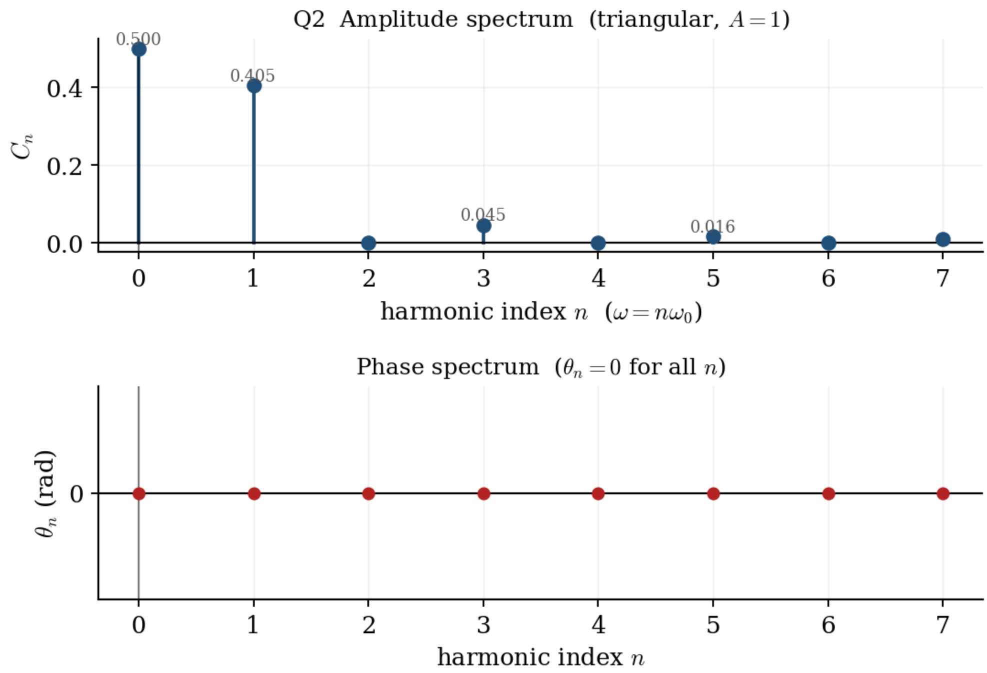

Figure 2: Q2: only odd harmonics are present, decaying as $1/n^2$; all phases are zero.

1.3 Q3 — Exponential FS of an odd square wave

Reading of the figure

The sketch is read as an **odd square pulse** of period $T_0 = 3$ ($\omega_0 = \frac{2\pi}{3}$): one period equals $+1$ on $(-1, 0)$, -1 on $(0, 1)$, and 0 on the remaining strips. If your figure has different levels/widths, only the constants below scale.

$$D_0 = \frac{1}{3}\left[\int_{-1}^{0}(+1)\,\mathrm{d}t + \int_{0}^{1}(-1)\,\mathrm{d}t\right] = 0.$$

For $n \neq 0$,

$$D_n = \frac{1}{3}\left[\int_{-1}^{0}\mathrm{e}^{-\mathrm{j}n\omega_0 t}\,\mathrm{d}t - \int_{0}^{1}\mathrm{e}^{-\mathrm{j}n\omega_0 t}\,\mathrm{d}t\right] = \frac{1}{3}\cdot\frac{2-\mathrm{e}^{\mathrm{j}n\omega_0}-\mathrm{e}^{-\mathrm{j}n\omega_0}}{-\mathrm{j}n\omega_0} = \frac{2\left(1-\cos\frac{2\pi n}{3}\right)}{-\mathrm{j}\,2\pi n} = \mathrm{j}\,\frac{1-\cos\frac{2\pi n}{3}}{\pi n}.$$

Since $\cos\frac{2\pi n}{3} = 1$ when $3 \mid n$ (giving $D_n = 0$) and $= -\frac{1}{2}$ otherwise,

Result

$$D_n = \begin{cases} \dfrac{3\mathrm{j}}{2\pi n}, & 3 \nmid n, \\ 0, & 3 \mid n, \end{cases} \qquad |D_n| = \frac{3}{2\pi|n|}, \qquad \angle D_n = \begin{cases} +\pi/2, & n > 0, \\ -\pi/2, & n < 0. \end{cases}$$

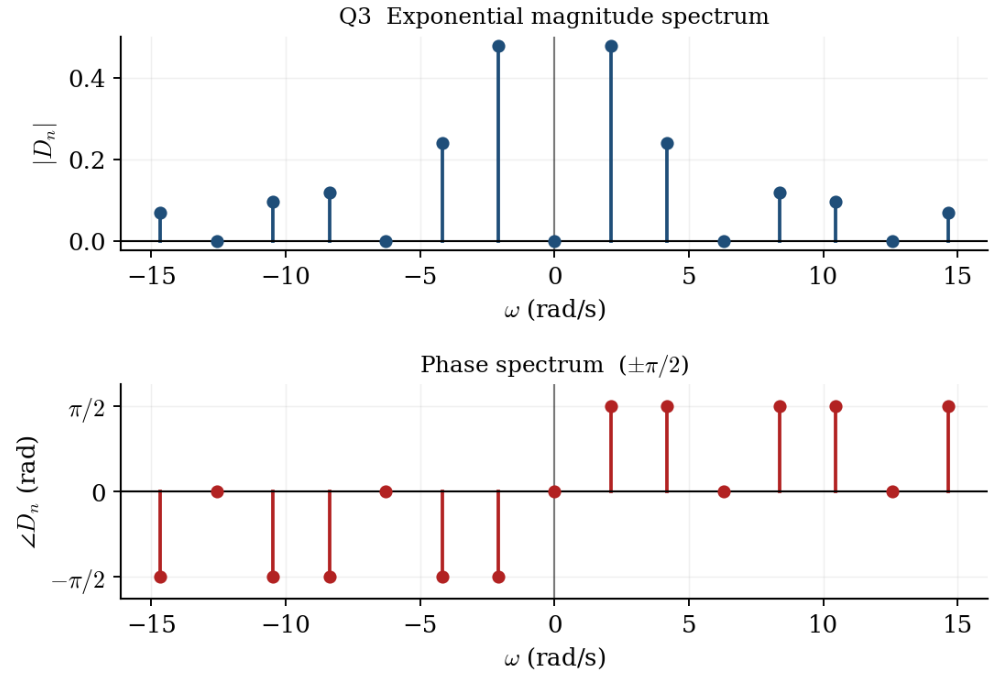

Figure 3: Q3: purely imaginary coefficients — magnitude $\propto 1/|n|$ with every third line missing, phase $\pm\pi/2$.

1.4 Q4 — Exponential FS of a sawtooth

Reading of the figure

The ramp is read as $x(t) = \frac{2}{3}t$ on $[0, 3]$, repeated with period $T_0 = 3$ ($\omega_0 = \frac{2\pi}{3}$).

$$D_0 = \frac{1}{3}\int_0^3 \tfrac{2}{3}t\,\mathrm{d}t = 1.$$

With $a = -\mathrm{j}n\omega_0$ and $\int t\,\mathrm{e}^{at}\,\mathrm{d}t = \dfrac{\mathrm{e}^{at}}{a^2}(at-1)$, and noting $\omega_0 \cdot 3 = 2\pi \Rightarrow \mathrm{e}^{a\cdot 3} = \mathrm{e}^{-\mathrm{j}2\pi n} = 1$,

$$D_n = \frac{2}{9}\int_0^3 t\,\mathrm{e}^{-\mathrm{j}n\omega_0 t}\,\mathrm{d}t = \frac{2}{9}\cdot\frac{-\mathrm{j}2\pi n}{(-\mathrm{j}n\omega_0)^2} = \frac{2}{9}\cdot\frac{9\mathrm{j}}{2\pi n} = \frac{\mathrm{j}}{\pi n}.$$

Result

$$D_0 = 1, \qquad D_n = \frac{\mathrm{j}}{\pi n}\ (n \neq 0), \qquad |D_n| = \frac{1}{\pi|n|}, \qquad \angle D_n = \pm\frac{\pi}{2}.$$

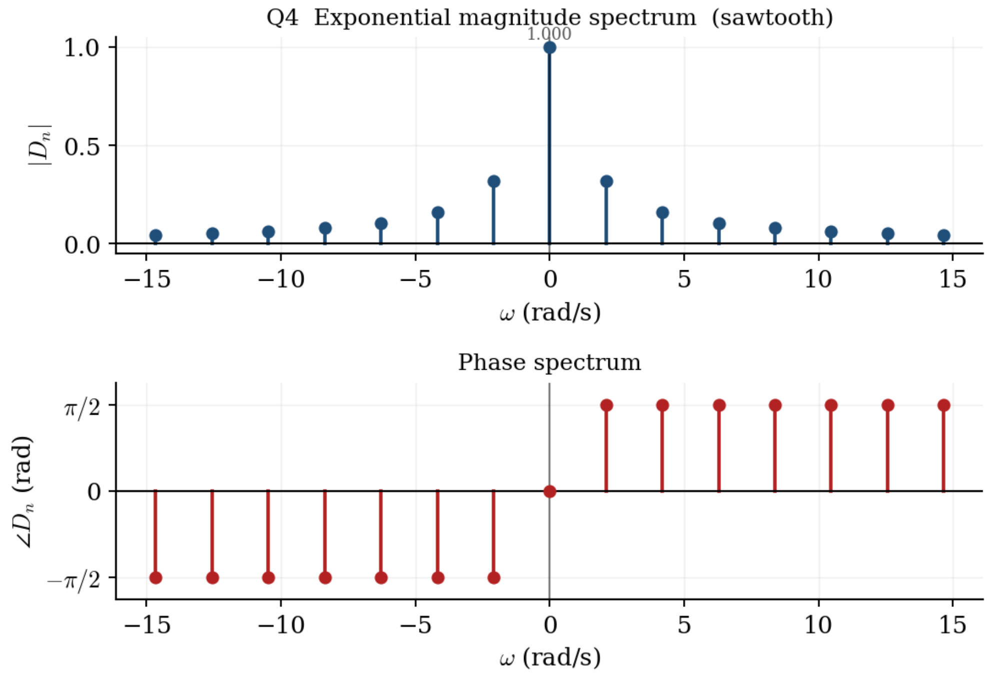

Figure 4: Q4: sawtooth spectrum — a DC value of 1, magnitudes $\propto 1/|n|$, phase $\pm\pi/2$.

1.5 Q5 — Compact-trig FS of an impulse train

Here $x(t) = 2\pi \sum_k \delta(t-2k)$, so $T_0 = 2$ and $\omega_0 = \pi$. Over the one period $(-1, 1)$ only $\delta(t)$ lies inside, hence

$$D_n = \frac{1}{2}\int_{-1}^{1} 2\pi\,\delta(t)\,\mathrm{e}^{-\mathrm{j}n\pi t}\,\mathrm{d}t = \pi \quad \text{for every } n.$$

A flat exponential spectrum gives $C_0 = \pi$ and $C_n = 2|D_n| = 2\pi$, with $\theta_n = 0$.

Result

$$x(t) = \pi + 2\pi\big[\cos\pi t + \cos 2\pi t + \cos 3\pi t + \cdots\big].$$

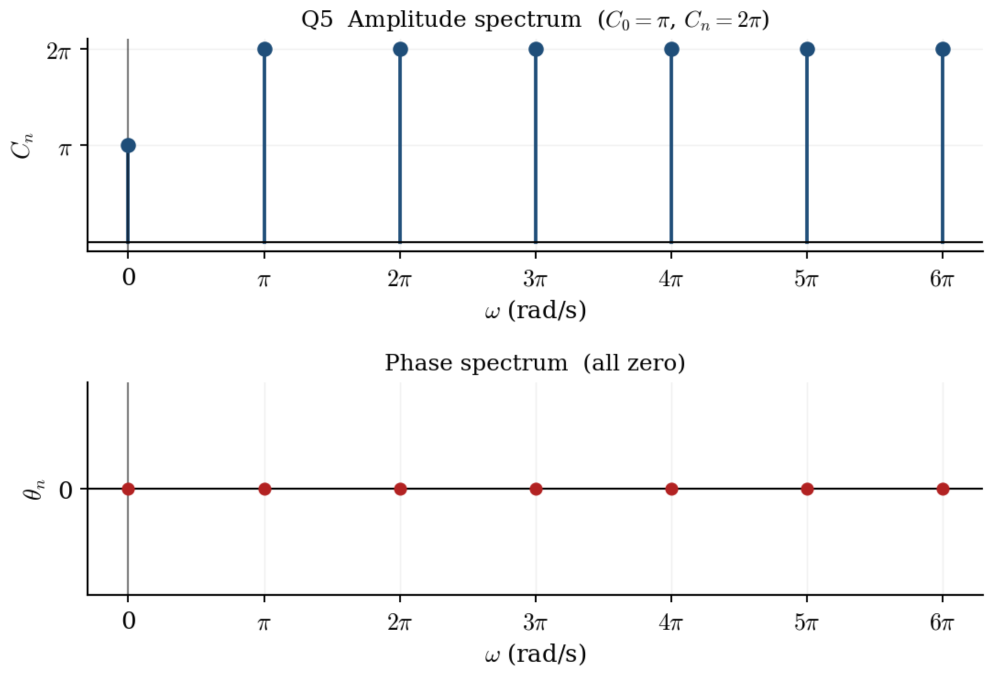

Figure 5: Q5: an impulse train has a flat spectrum — $C_0 = \pi$, $C_n = 2\pi$, all phases zero.

1.6 Q6 — Exponential FS of a weighted impulse train

Reading of the figure

Impulses sit at every integer with weight $+1$ at odd integers and -2 at even integers; period $T_0 = 2$, $\omega_0 = \pi$. Inside one period only $t = 0$ (weight -2) and $t = 1$ (weight $+1$) occur.

$$D_n = \frac{1}{2}\Big[(-2)\mathrm{e}^0 + (1)\mathrm{e}^{-\mathrm{j}n\pi}\Big] = \frac{-2 + (-1)^n}{2} = \begin{cases} -\frac{1}{2}, & n \text{ even}, \\ -\frac{3}{2}, & n \text{ odd}. \end{cases}$$

All coefficients are negative real numbers, so every phase equals π.

Result

$$D_0 = -\tfrac{1}{2}, \qquad |D_n| = \begin{cases} \frac{1}{2}, & n \text{ even} \\ \frac{3}{2}, & n \text{ odd} \end{cases}, \qquad \angle D_n = \pi \text{ for all } n.$$

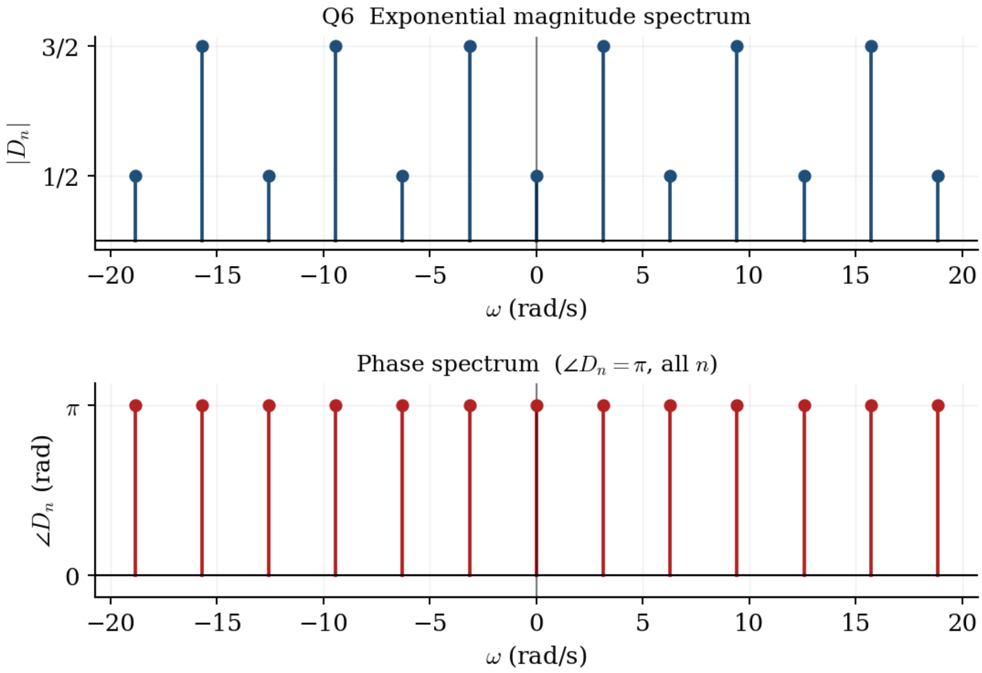

Figure 6: Q6: even/odd harmonics take magnitudes $\frac{1}{2}$ and $\frac{3}{2}$; the constant negative sign shows as a uniform phase of π.

1.7 Q7 — Periodicity and harmonics

A sum of sinusoids is periodic *iff* every pairwise ratio of frequencies is rational; the fundamental ω_0 is then the largest number of which all frequencies are integer multiples.

(a) $2\sin 3t + 7\cos \pi t$: ratio $3/\pi$ is **irrational** $\Rightarrow$ **aperiodic**.

(b) $7\cos \pi t + 5\sin 2\pi t$: frequencies $\pi, 2\pi$; $\omega_0 = \pi$, $T_0 = 2$. **Periodic**; these are the 1$^{\text{st}}$ and 2$^{\text{nd}}$ harmonics.

(c) $3\cos\sqrt{2}\,t + 5\cos 2t$: ratio $\sqrt{2}/2$ is **irrational** $\Rightarrow$ **aperiodic**.

(d) $\sin\frac{5t}{2} + 3\cos 3t + 3\sin\left(\frac{t}{7} + 30°\right)$: frequencies $\frac{5}{2}, 3, \frac{1}{7}$. The largest common divisor is $\omega_0 = \frac{1}{14}$, so $T_0 = 28\pi$. **Periodic**; the terms are the 35$^{\text{th}}$ ($\frac{5}{2}$), 42$^{\text{nd}}$ (3) and 2$^{\text{nd}}$ ($\frac{1}{7}$) harmonics.

1.8 Q8 — Compact-trig and exponential FS of a given signal

$$f(t) = 3 + \sqrt{3}\cos 2t + \sin 2t + \sin(5t - \tfrac{\pi}{6}) - \tfrac{1}{2}\cos(7t + \tfrac{\pi}{3}), \qquad \omega_0 = 1.$$

(a) Compact trigonometric form. Combine the two $\omega = 2$ terms and convert each sine to a cosine:

$$\sqrt{3}\cos 2t + \sin 2t = 2\cos(2t - 30°), \quad \sin(5t - \tfrac{\pi}{6}) = \cos(5t - \tfrac{2\pi}{3}), \quad -\tfrac{1}{2}\cos(7t + \tfrac{\pi}{3}) = \tfrac{1}{2}\cos(7t - \tfrac{2\pi}{3}).$$

Result

$$f(t) = 3 + 2\cos(2t - 30°) + \cos(5t - 120°) + \tfrac{1}{2}\cos(7t - 120°).$$
$$C_0 = 3, \quad C_2 = 2\,(\theta = -30°), \quad C_5 = 1\,(\theta = -120°), \quad C_7 = \tfrac{1}{2}\,(\theta = -120°).$$

(b) Exponential form. Using $C_n = 2|D_n|$ and $\angle D_{\pm n} = \mp\theta_n$:

$$D_0 = 3, \quad |D_{\pm 2}| = 1\,(\angle{=}\mp 30°), \quad |D_{\pm 5}| = \tfrac{1}{2}\,(\angle{=}\mp 120°), \quad |D_{\pm 7}| = \tfrac{1}{4}\,(\angle{=}\mp 120°).$$

(c, d) Equivalence. Each conjugate pair recombines as $D_n e^{jnt} + D_{-n}e^{-jnt} = 2|D_n|\cos(nt + \angle D_n) = C_n\cos(nt+\theta_n)$, which reproduces the compact-trig terms above and hence the original $f(t)$.

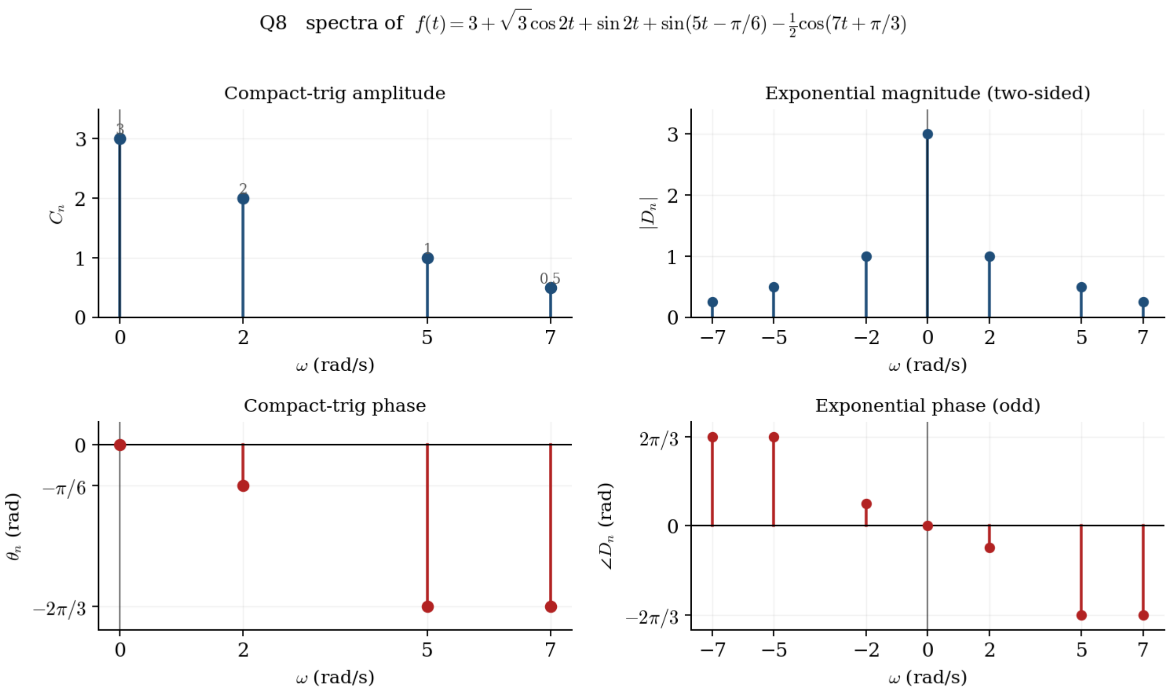

Figure 7: Q8: compact-trig (one-sided, left) and exponential (two-sided, right) spectra. The exponential phase is the odd reflection of the trig phase.

1.9 Q9 — A signal given in exponential form

$$x(t) = (2 + j2)e^{-j3t} + j2\,e^{-jt} + 3 - j2\,e^{jt} + (2 - j2)e^{j3t}, \qquad \omega_0 = 1.$$

Matching to $\sum_n D_n e^{jnt}$ reads the coefficients directly:

$$D_0 = 3, \quad D_1 = -j2, \quad D_{-1} = j2, \quad D_3 = 2 - j2, \quad D_{-3} = 2 + j2.$$

Since $D_{-1} = D_1^*$ and $D_{-3} = D_3^*$, the signal is **real**. Hence

$$|D_1| = 2\,(\angle{-90^\circ}),\ |D_{-1}| = 2\,(\angle{+90^\circ}), \quad |D_3| = 2\sqrt{2}\,(\angle{-45^\circ}),\ |D_{-3}| = 2\sqrt{2}\,(\angle{+45^\circ}).$$

Compact trig. $C_0 = 3$, $C_1 = 2|D_1| = 4$ $(\theta_1 = -90^\circ)$, $C_3 = 2|D_3| = 4\sqrt{2}$ $(\theta_3 = -45^\circ)$:

Result

$$x(t) = 3 + 4\cos(t - 90^\circ) + 4\sqrt{2}\cos(3t - 45^\circ) = 3 + 4\sin t + 4\cos 3t + 4\sin 3t.$$

(d) Bandwidth $= \omega_{\max} - \omega_{\min} = 3 - 1 = 2$ rad/s (highest harmonic present is 3 rad/s).

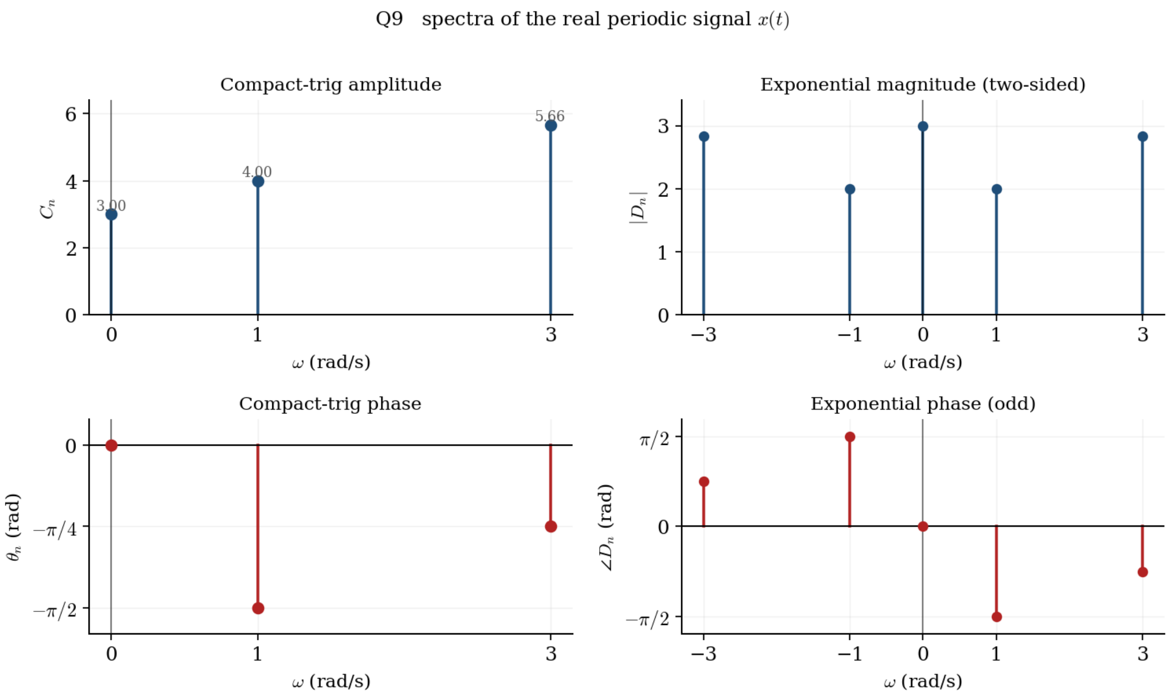

Figure 8: Q9: two-sided exponential spectrum (right) and the equivalent one-sided compact-trig spectrum (left).

1.10 Q10 — From a given exponential spectrum to the signal

Reading of the figure

The plotted lines are read as $|D_0| = 2$, $|D_{\pm1}| = 1.5$, $|D_{\pm2}| = 1$, $|D_{\pm3}| = 0.5$ (a triangular envelope reaching zero at $n = \pm4$), with phase $\angle D_n = -n\pi/3$ and $\omega_0 = 1$ rad/s.

(a) Exponential synthesis. The $n = \pm3$ terms carry phase $\mp\pi$, i.e. $\mathrm{e}^{\mp \mathrm{j}\pi} = -1$:

$$x(t) = 2 + 1.5\,\mathrm{e}^{-\mathrm{j}\pi/3}\mathrm{e}^{\mathrm{j}t} + 1.5\,\mathrm{e}^{\mathrm{j}\pi/3}\mathrm{e}^{-\mathrm{j}t} + \mathrm{e}^{-\mathrm{j}2\pi/3}\mathrm{e}^{\mathrm{j}2t} + \mathrm{e}^{\mathrm{j}2\pi/3}\mathrm{e}^{-\mathrm{j}2t} - 0.5\,\mathrm{e}^{\mathrm{j}3t} - 0.5\,\mathrm{e}^{-\mathrm{j}3t}.$$

(b) Compact trig. $C_0 = 2$, $C_n = 2|D_n|$, $\theta_n = \angle D_n$:

$$C_1 = 3\,(\theta = -60^\circ), \quad C_2 = 2\,(\theta = -120^\circ), \quad C_3 = 1\,(\theta = -180^\circ).$$

Result

$$x(t) = 2 + 3\cos(t - 60^\circ) + 2\cos(2t - 120^\circ) - \cos 3t.$$

(c, d) Pairing $D_n\mathrm{e}^{\mathrm{j}nt} + D_{-n}\mathrm{e}^{-\mathrm{j}nt} = C_n\cos(nt + \theta_n)$ confirms the two representations are identical.

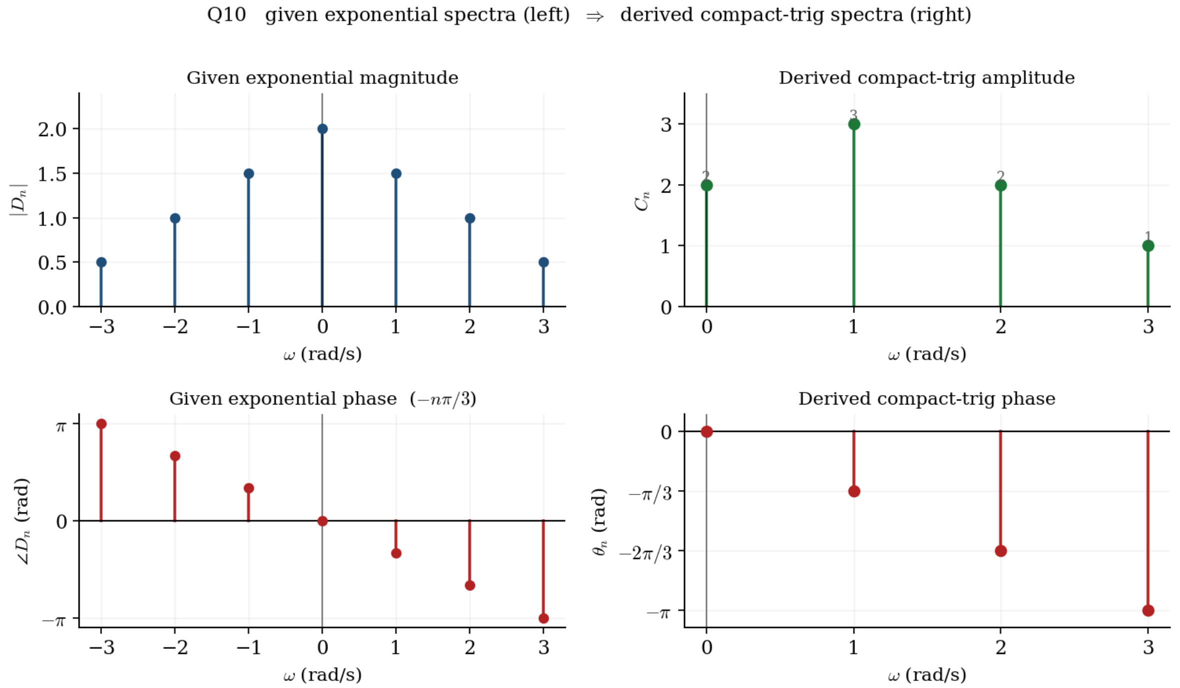

Figure 9: Q10: the given exponential spectra (blue/red, left) and the derived compact-trig spectra (green/red, right).

2 Assignment 4 — Fourier Transform

2.1 Q1 — Convolution $y = x_1 * x_2$

$x_1(t) = \text{sinc}(4\pi t)$, $x_2(t) = \cos 2\pi t$.

(a) With $a = 4\pi$ in the sinc pair,

$$X_1(\omega) = \frac{\pi}{4\pi}\,\text{rect}\left(\frac{\omega}{8\pi}\right) = \frac{1}{4} \quad \text{for } |\omega| < 4\pi,$$

a gate of height $\frac{1}{4}$. (The time signal has zeros at $t = k/4$ and unit peak.)

(b) $X_2(\omega) = \pi\big[\delta(\omega - 2\pi) + \delta(\omega + 2\pi)\big]$.

(c) Convolution in time ⇒ multiplication in frequency. Because $|\pm 2\pi| < 4\pi$, the gate equals $\frac{1}{4}$ at both impulses:

$$Y(\omega) = X_1(\omega)X_2(\omega) = \frac{\pi}{4}\big[\delta(\omega - 2\pi) + \delta(\omega + 2\pi)\big] \implies \boxed{y(t) = \tfrac{1}{4}\cos 2\pi t}.$$

Intuitively, convolving with a sinc is ideal low-pass filtering; the 2π tone sits in the passband and is merely scaled by $\frac{1}{4}$.

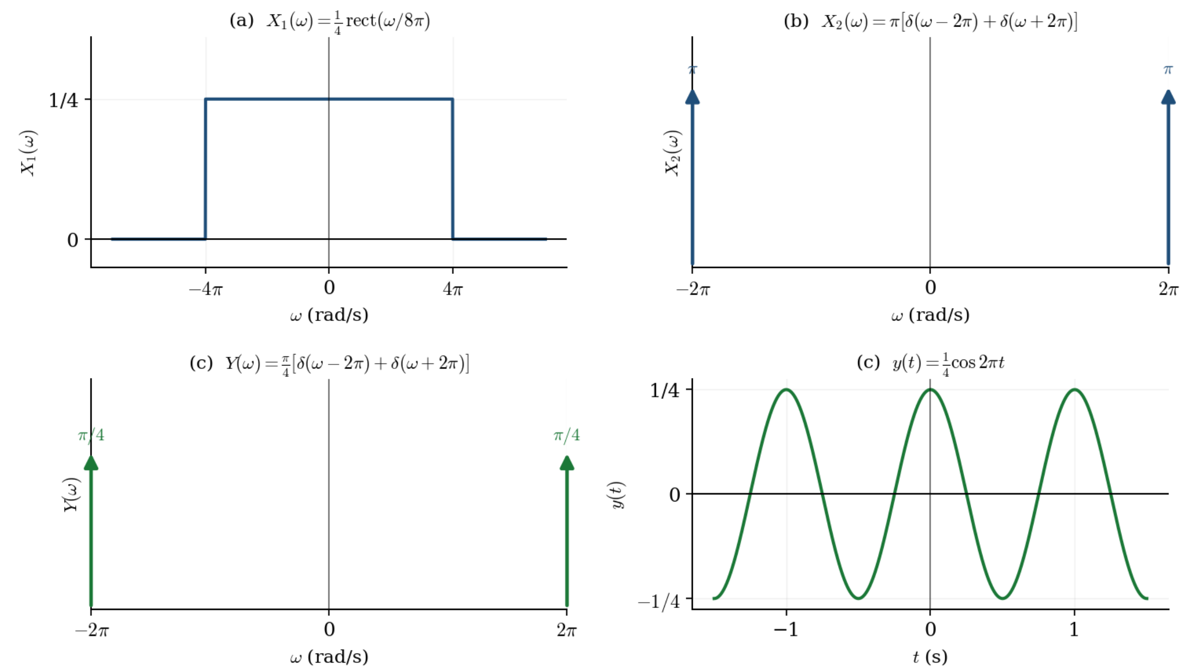

Figure 10: Q1: the sinc gate X_1, the cosine's impulses X_2, their product Y, and the resulting time signal $y(t) = \frac{1}{4}\cos 2\pi t$.

2.2 Q2 — Property-based transforms of a triangular spectrum

The given spectrum is the real triangle $X(\omega) = 1 - \dfrac{2|\omega|}{\pi}$ for $|\omega| \le \dfrac{\pi}{2}$ (zero phase).

(i) **Time scaling** $x(3t) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{3}X(\omega/3)$: peak shrinks to $\frac{1}{3}$, base widens to $\pm\frac{3\pi}{2}$; phase still 0.

(ii) **Time shift** $x(t-5) \overset{\mathcal{F}}{\longleftrightarrow} \mathrm{e}^{-\mathrm{j}5\omega}X(\omega)$: magnitude unchanged; a *linear* phase -5ω is added.

(iii) **Modulation** $x(t)\cos 2\pi t \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2}\big[X(\omega-2\pi)+X(\omega+2\pi)\big]$: two half-height triangles centred at $\pm 2\pi$.

(iv) **Convolution** $x(t) * \mathrm{sinc}(\frac{\pi t}{4})$. Since $\mathrm{sinc}(\frac{\pi}{4}t) \overset{\mathcal{F}}{\longleftrightarrow} 4\,\mathrm{rect}(\frac{\omega}{\pi/2})$ (height 4 for $|\omega| < \frac{\pi}{4}$), the result is $4X(\omega)$ restricted to $|\omega| < \frac{\pi}{4}$: value 4 at $\omega = 0$, falling linearly to 2 at $\pm\frac{\pi}{4}$, then 0.

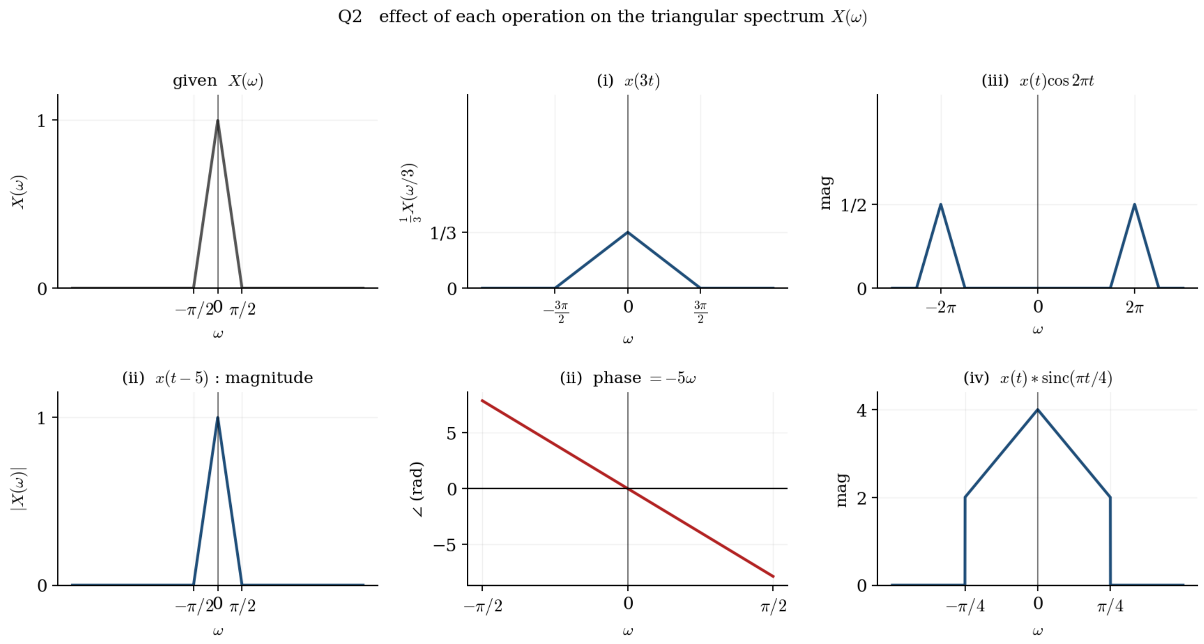

Figure 11: Q2: each Fourier property reshapes the base triangle $X(\omega)$ — scaling, a linear-phase shift, modulation to $\pm 2\pi$, and band-limiting by convolution.

2.3 Q3 — Recovering $x(t)$ by frequency shifting

$X(\omega)$ consists of two unit gates on $[3, 5]$ and $[-5, -3]$. Write them as a baseband gate $G(\omega) = \mathrm{rect}(\frac{\omega}{2})$ (i.e. 1 for $|\omega| < 1$) shifted by ± 4:

$$X(\omega) = G(\omega - 4) + G(\omega + 4).$$

The baseband gate inverts to $g(t) = \dfrac{1}{2\pi}\displaystyle\int_{-1}^{1} \mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega = \dfrac{\sin t}{\pi t} = \dfrac{1}{\pi}\,\mathrm{sinc}(t)$. Frequency shifting $(G(\omega \mp 4) \xleftrightarrow{\mathcal{F}} g(t)\mathrm{e}^{\pm \mathrm{j}4t})$ then gives

Result

$$x(t) = g(t)\left(\mathrm{e}^{\mathrm{j}4t} + \mathrm{e}^{-\mathrm{j}4t}\right) = \frac{2}{\pi}\,\mathrm{sinc}(t)\cos 4t = \frac{2\sin t}{\pi t}\cos 4t.$$

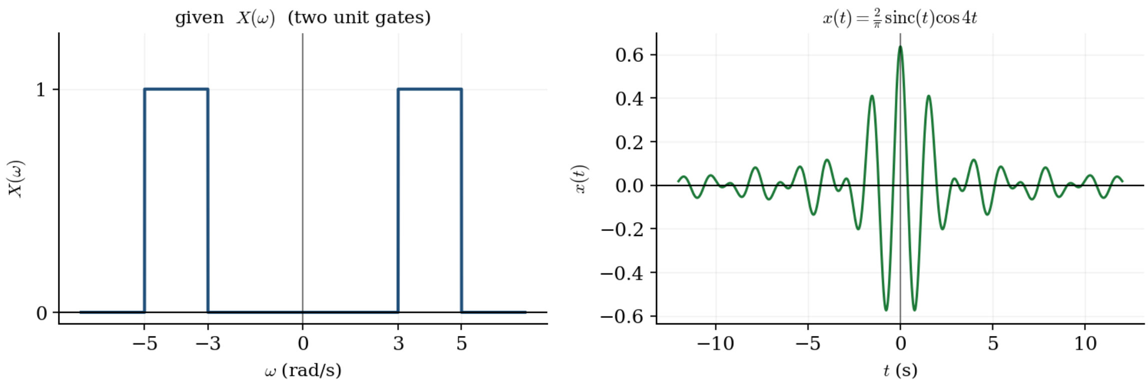

Figure 12: Q3: the two-gate spectrum (left) and the reconstructed signal — a sinc envelope modulated by $\cos 4t$ (right).

2.4 Q4 — Inverse transform by the convolution property

$$X(\omega) = \frac{1}{(a+\mathrm{j}\omega)^2} = \left[\frac{1}{a+\mathrm{j}\omega}\right]^2.$$

Because $\dfrac{1}{a+\mathrm{j}\omega} \overset{\mathcal{F}}{\longleftrightarrow} \mathrm{e}^{-at}u(t)$, a product in frequency is a convolution in time, so $x(t)$ is $e^{-at}u(t)$ convolved with itself:

$$x(t) = \int_0^t \mathrm{e}^{-a\tau}\mathrm{e}^{-a(t-\tau)}\,\mathrm{d}\tau = \mathrm{e}^{-at}\int_0^t \mathrm{d}\tau = t\,\mathrm{e}^{-at}, \quad t \geq 0.$$

Result

$$x(t) = t\,\mathrm{e}^{-at}\,u(t).$$

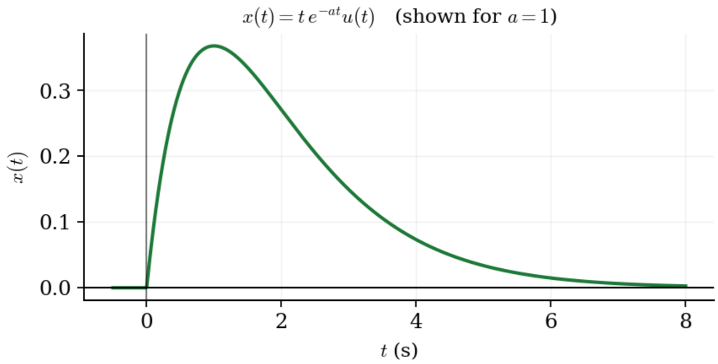

Figure 13: Q4: the self-convolution of a decaying exponential rises linearly then decays — the classic $t\,e^{-at}u(t)$ shape.

2.5 Q5 — A truncated exponential and three constructions

The base signal is $x(t) = \mathrm{e}^{-2t}$ for $0 \leq t \leq 1$ (zero elsewhere), with

$$X(\omega) = \int_0^1 \mathrm{e}^{-(2+\mathrm{j}\omega)t}\,\mathrm{d}t = \frac{1-\mathrm{e}^{-2}\mathrm{e}^{-\mathrm{j}\omega}}{2+\mathrm{j}\omega}.$$

Reading of the figure

The three companion sketches are read as the even extension, the odd extension, and a time-expansion of the same pulse. Each is handled by one Fourier property, so any alternative reading simply selects a different property from the same list.

- $x_1(t) = x(t) + x(-t)$ (**even**): $X_1(\omega) = X(\omega) + X(-\omega) = 2\,\mathrm{Re}\{X(\omega)\}$.
- $x_2(t) = x(t) - x(-t)$ (**odd**): $X_2(\omega) = X(\omega) - X(-\omega) = 2\mathrm{j}\,\mathrm{Im}\{X(\omega)\}$.
- $x_3(t) = x(t/2)$ (**time-expanded**): $X_3(\omega) = 2\,X(2\omega)$.

The even part is purely real, the odd part purely imaginary, and the stretched pulse has a correspondingly compressed, twice-as-tall spectrum.

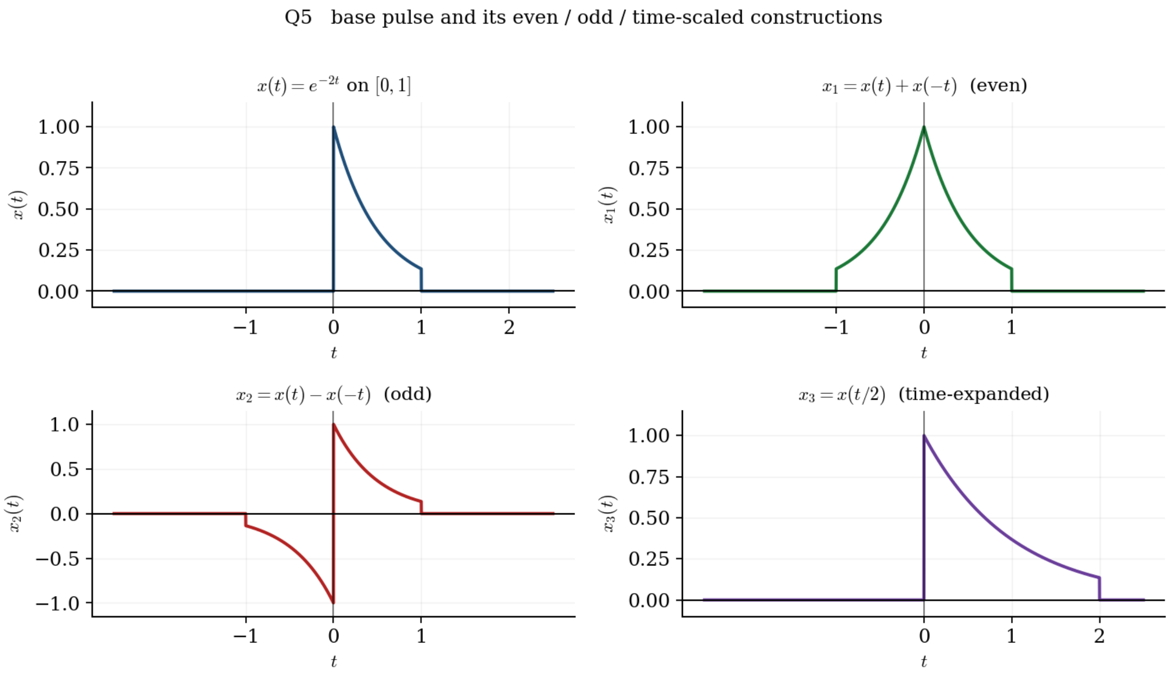

Figure 14: Q5: the base pulse and its even, odd, and time-expanded ($t \to t/2$) constructions in the time domain.

End of solutions. Orange boxes mark every place where a figure was interpreted — verify those readings against your own printed assignment.